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Optimal Allocation of Risk-Reduction Resources in Event Trees

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In this paper, we present a novel quantitative analysis for the strategic planning decision problem of allocating certain available prevention and protection resources to, respectively, reduce the failure probabilities of system safety measures and the total expected loss from a sequence of events. Using an *event tree optimization* approach, the resulting risk-reduction scenario problem is modeled and then reformulated as a specially structured nonconvex factorable program. We derive a tight linear programming relaxation along with related theoretical insights that serve to lay the foundation for designing a tailored branch-and-bound algorithm that is proven to converge to a global optimum. Computational experience is reported for a hypothetical case study, as well as for several realistic simulated test cases, based on different parameter settings. The results on the simulated test cases demonstrate that the proposed approach dominates the commercial software BARON v7.5 when the latter is applied to solve the original model by more robustly yielding provable optimal solutions that are at an average of 16.6% better in terms of objective function value; and it performs competitively when both models are used to solve the reformulated problem, particularly for larger test instances.

Key words: risk management; risk reduction; event trees; system safety; global optimization; factorable programming; branch-and-bound
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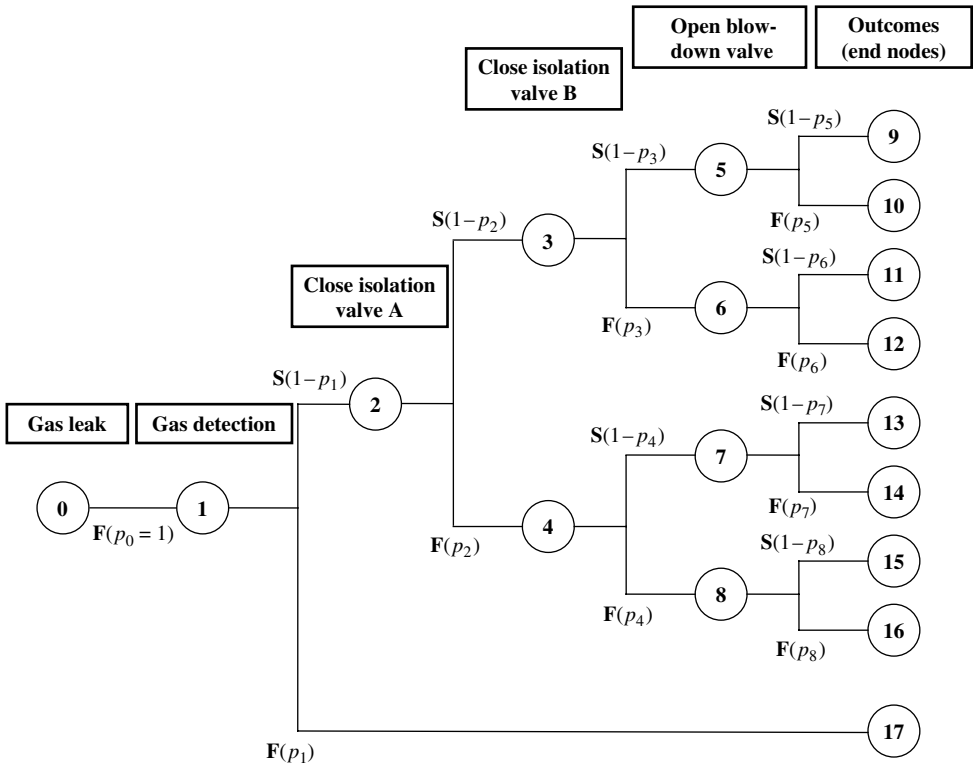
1. Introduction

One of the primary tasks of risk management relates to the issue of minimizing the loss that might occur as a result of a series of hazardous events. Among the methods most commonly used to solve such uncertainty-related problems is *decision tree* analysis. In particular, two types of decision trees, namely, *fault trees* and *event trees*, are widely used in the practice of *probabilistic risk assessment* (see Kumamoto and Henley 2000). The principal difference between fault trees and event trees is that the basis for fault trees lies in *deductive* (backward) logic, whereas event trees usually conform to *inductive* (forward) logic. The main focus of this paper is in analyzing and optimizing event trees, which is described below and illustrated by a specific application.

An event tree begins with an *initiating event point*, represented as node zero, which could either be a component failure within the system under study or the result of an external hazardous occurrence. Beginning with this situation, a chain of successive events is triggered in a cascading fashion, where each event (branch) represents the success or failure of some safety measure applied at a particular node. Note that a given safety measure can be associated with several possible nodes. Also, for simplicity, we consider a single initiating event point and a pair of binary states (success or failure) for each event (i.e., Bernoulli events), but our approach could likewise be applied to generic trees having multiple initiating occurrences and an m-state mechanism.

Figure 1 displays an event tree for a gasline rupture situation related to an offshore oil and gas production platform (see Andrews and Dunnett 2000). Here, at each node, the application of a certain safety measure (e.g., closure of isolation valves to localize the damage or opening of blowdown valves to depressurize certain sections) can lead to one of two immediate scenarios based on the success or failure of this measure. Each resulting chain of cascading events terminates in

Figure 1 A Gas Leak Event Tree



some end node of the tree, which entails a specific *consequence* (loss). The *expected loss* or *risk* associated with an event tree is the sum of the probability-weighted consequences over all the chains in the tree. Note that for independent events, the probability of occurrence of a particular final outcome is the product of the probabilities associated with the links that lie on the unique chain connecting the initiating event point to the corresponding end node. Alternatively, in lieu of assuming independence, the same generic formula is applicable by viewing each event probability as a *conditional probability*, given the sequence of events on the unique chain leading up to it.

The use of event trees to analyze system failures began in the 1970s, when the U.S. Nuclear Regulatory Commission performed risk assessment tests in nuclear power plants (see Rasmussen 1975). Since then, event tree analysis has been used to study system risk in various contexts arising in both the public and private sectors. These studies include steam generator tube ruptures (Zhang and Yan 1999), water resource planning (Beim and Hobbs 1997), fusion-fission hybrid reactor failures (Yang and Qiu 1993), electrical accident countermeasure systems for mines (Collins and Cooley 1983), failure of temporary structures (Hadipriono et al. 1986), reliability analysis of high-voltage transmission systems (Ohba et al. 1984), system safety related to computer security (Leveson 1995), and emergency response in the context of chemical hazards or spills (Raman 2004 and Zhang et al. 2004). All these case studies deal mainly with the development of the event tree and the derivation of the associated (conditional) probabilities and consequences as prompted by the specific application, along with the computation of the risk or expected consequence (see Leveson 1995). Takaragi et al. (1983) utilize minimal cut or prime implicant sets for eliminating certain events from the event tree and providing an upper-bounding approximation for the failure probabilities. Based on this work, several modifications and extensions have appeared in the literature. Using mostly binary decision diagrams, Sinnamon and Andrews (1996; 1997a, b), Rauzy (1993, 1996), and Andrews and Dunnett (2000) have proposed several quantitative and qualitative approaches for calculating the (conditional) link probabilities and the total expected loss in fault trees and event trees, particularly for those cases when the events are not independent. By viewing event trees in terms of transition matrices that evolve from an entry state to various exit states, Kaplan (1982) initiated matrix theory formalisms for event tree analysis and provided related computational insights. Also, Unwin (1984) has presented a compact numerical representation for the different scenarios that occur in an event tree in order to reduce computer memory requirements. In addition, certain fuzzy set-based event tree analyses have been

reported in Huang et al. (2001), Kenarangui (1991), Jin et al. (2003), and Patra et al. (1995).

Recently, two important studies of risk-related resource allocation have appeared in the literature, one in the context of space exploration and the other in relation to homeland security. The first one is the risk and uncertainty based integrated concurrent (RUBIC) design methodology of Mehr and Tumer (2006), which is essentially a portfolio optimization procedure that reduces the overall system risk by examining the impact of allocated resources on multiple elements in the system. The second one is a treatise on critical infrastructure protection by Lewis (2006), which presents a heuristic ranking method based on an organizing principle algorithm using a single resource constraint to represent a budgetary restriction.

In this paper, we introduce a novel decision support framework for strategic planning that can help managers prepare for and manage emergency situations to reduce the overall risk associated with a given hazardous scenario. For example, in planning to reduce the risk associated with a gas leak, certain available preventive or protective *event-related resources* such as investments in improved technologies or equipment could be used to reduce the failure likelihood of the different safety features, such as leak detection, closing of critical valves, etc., given that a particular sequence of preceding events has occurred (see Figure 1). Likewise, investments in certain available *consequence-related resources* such as clean-up devices and trained emergency response personnel could be brought to bear to ameliorate the potential losses associated with the end nodes of the decision tree. In such a situation, the goal would be to achieve the most effective deployment of the limited available resources by manipulating the event (conditional) probabilities and the resultant consequences associated with the event tree in order to minimize the overall risk.

Accordingly, let $i \in \{1, \ldots, I\}$ index the set of nodes (*event points*) at which some safety feature is deployed in a binary event tree (e.g., $I = 8$ in Figure 1) leading to consequent success or failure events. Let p_i and $(1 - p_i)$, respectively, denote the conditional probabilities of failure and success associated with the outcomes resulting from the application of the particular safety feature at node or event point i, given that the sequence of events leading from the root node 0 to node i has occurred. Now, suppose that we are given a set of *event-related resources* $m \in \{1, \ldots, M\}$ that can be applied to control the outcome probabilities. Correspondingly, let q_{im} be the quantity of event-related resource m that is allocated to reduce the associated conditional (failure) probability p_i of the safety feature applied at node i. To accommodate various special cases where we might have certain particular investment decisions that relate to the q_{im}-variables, we define a general constraining polyhedral set $\mathcal{P}$ in our model, described in §2, to handle such situations.

Using standard logit model theory, we assume that the logit function for p_i is linearly related to the allocations of the prevention resources, i.e.,

$$\ln\left[\frac{p_i}{1-p_i}\right] = a_{i0} - \sum_{m=1}^{M} a_{im} q_{im}, \quad \forall\, i = 1, \ldots, I,$$

where $a_{i0}, \ldots, a_{iM}$ are nonnegative constants associated with the logit choice model such that $0 < p_i < 1$, $\forall\, i = 1, \ldots, I$. Moreover, let the set of final consequences (end nodes of the event tree) be indexed by the set $j \in \tau$, and let l_j denote the loss magnitude associated with end node j ($\tau \equiv \{9, \ldots, 17\}$ in Figure 1). As before, suppose that we are given a set of *consequence-related resources* $n \in \{1, \ldots, N\}$, and let r_{jn} be the quantity of consequence-related resource n that is allocated to reduce the loss magnitude l_j of consequence j. Similar to the case of event-related resources, special additional restrictions on the r_{jn}-variables can be accommodated within the aforementioned set $\mathcal{P}$. We assume that the magnitude of consequence j is linearly related to the allocation of such resources, i.e.,

$$l_j = b_{j0} - \sum_{n=1}^{N} b_{jn} r_{jn}, \quad \forall\, j \in \tau,$$

where $b_{j0}, \ldots, b_{jN}$ are given nonnegative constants. (Alternatively, we could consider a diminishing marginal return loss function of the type $l_j = b_{j0} \exp\{-h_j \sum_{n=1}^{N} b_{jn} r_{jn}\}$, $\forall\, j \in \tau$, having nonnegative parameter values. The algorithm proposed in an intended sequel can be identically applied for such loss functions upon taking logarithms.)

Furthermore, let s_m and t_n denote the magnitudes of the total available units of event-related resource m and consequence-related resource n, respectively. Finally, let c_{im}, d_{jn}, and β represent the per-unit cost of allocating event-related resource m to node i to reduce p_i, the per-unit cost of allocating consequence-related resource n to end node j to reduce l_j, and the total available budget, respectively. The *event tree optimization* (ETO) problem is then concerned with determining an optimal allocation of risk reduction resources so as to minimize the total expected loss, subject to the resource availability and budgetary restrictions.

The remainder of this paper is organized as follows. In §2, we formulate a mathematical model for the ETO problem and then develop a suitable reformulation along with a tight lower-bounding representation through some transformations and polyhedral approximations. Section 3 describes a specialized branch-and-bound procedure to solve the

formulated problem and establishes its convergence. Some computational experience is presented in §4 using a hypothetical case study as well as several realistic simulated test cases. Finally, §5 concludes the paper with a summary and directions for future research.

2. Event Tree Optimization Model

Consider an event tree T rooted at the initiating event node 0, and let τ denote the set of terminal or end nodes j, each associated with the corresponding loss l_j. For each $j \in \tau$, let $\mathscr{L}_j \neq \varnothing$ denote the set of nodes k lying on the unique chain from the root node 0 to node j, excluding nodes 0 and j. Note that for each $k \in \mathscr{L}_j$ there exists an associated link L_{jk}, say, that lies on the chain from node 0 to node j and connects node k to one of its immediate successor nodes in T that belongs to this chain. Accordingly, let

$$S_{1j} = \{k \in \mathscr{L}_j\text{: the (conditional) probability associated with link } L_{jk} \text{ is } p_k\}, \quad \forall j \in \tau, \quad \text{and}$$

$$S_{2j} = \{k \in \mathscr{L}_j\text{: the (conditional) probability associated with link } L_{jk} \text{ is } (1-p_k)\}, \quad \forall j \in \tau.$$

The ETO problem can then be formulated as follows, where the bounds specified for defining Ω in (1g) are assumed to be given (or implied by the scenario considerations; e.g., $l_j^u = b_{j0}$, $\forall j$, and $l_j^l = \max\{\varepsilon, b_{j0} - \sum_{n=1}^{N} b_{jn} t_n\}$, $\forall j$, where $\varepsilon > 0$). In either case, we assume throughout that the following program is feasible, as, for example, by virtue of the parameters and Ω being defined such that $(q, r) = (0, 0)$ induces a feasible solution.

ETO: Minimize
$$\sum_{j\in\tau} l_j \prod_{k\in S_{1j}} p_k \prod_{k\in S_{2j}} (1-p_k) \tag{1a}$$

subject to
$$\sum_{i=1}^{I} q_{im} \le s_m, \quad \forall m = 1, \ldots, M \tag{1b}$$

$$\sum_{j\in\tau} r_{jn} \le t_n, \quad \forall n = 1, \ldots, N \tag{1c}$$

$$\sum_{i=1}^{I}\sum_{m=1}^{M} c_{im} q_{im} + \sum_{j\in\tau}\sum_{n=1}^{N} d_{jn} r_{jn} \le \beta, \tag{1d}$$

$$\ln\left[\frac{p_i}{1-p_i}\right] = a_{i0} - \sum_{m=1}^{M} a_{im} q_{im}, \quad \forall i = 1, \ldots, I \tag{1e}$$

$$l_j = b_{j0} - \sum_{n=1}^{N} b_{jn} r_{jn}, \quad \forall j \in \tau \tag{1f}$$

$$0 < p_i^l \le p_i \le p_i^u < 1, \quad \forall i = 1, \ldots, I \tag{1g}$$

$$0 < l_j^l \le l_j \le l_j^u < \infty, \quad \forall j \in \tau \tag{1h}$$

$$(q, r) \in \mathscr{P}. \tag{1i}$$

In this formulation, the objective function (1a) seeks to minimize the total risk, constraints (1b) and (1c) impose the resource availability restrictions, (1d) enforces the budgetary limitation, (1e) and (1f) follow from the assumptions based on the logit choice model, (1g) and (1h), respectively, require the p- and l-variables to satisfy some specified bounding restrictions, and $\mathscr{P}$ in (1i) is a polyhedron (embedded in the nonnegative orthant) that includes any additional suitable constraints and variables as discussed before to further restrict the (q, r)-variables or to relate them to other decisions (in the simplest case, $\mathscr{P} = \{(q, r)\text{: } (q, r) \ge 0\}$). Observe that the nonconvexity in Problem ETO arises due to the polynomial objective function (1a) and the logarithmic term in (1e).

Let us now define the following auxiliary variables, along with their implied bounds, to conveniently reformulate Problem ETO. Denote

$$\theta_j = l_j \prod_{k\in S_{1j}} p_k \prod_{k\in S_{2j}} (1-p_k), \quad \forall j \in \tau. \tag{2a}$$

Then we have

$$\theta_j^l \le \theta_j \le \theta_j^u,$$

$$\text{where } \left\{ \begin{array}{l} \theta_j^l = l_j^l \displaystyle\prod_{k\in S_{1j}} p_k^l \prod_{k\in S_{2j}} (1-p_k^u) \\ \theta_j^u = l_j^u \displaystyle\prod_{k\in S_{1j}} p_k^u \prod_{k\in S_{2j}} (1-p_k^l) \end{array} \right\}, \quad \forall j \in \tau. \tag{2b}$$

Similarly, to linearize (1e) we introduce the variables

$$y_{1i} = \ln(p_i) \quad \text{and} \quad y_{2i} = \ln(1-p_i), \quad \forall i = 1, \ldots, I. \tag{3a}$$

Note that

$$y_{1i}^l \le y_{1i} \le y_{1i}^u \quad \text{and} \quad y_{2i}^l \le y_{2i} \le y_{2i}^u, \quad \text{where } y_{1i}^l = \ln(p_i^l),\ y_{1i}^u = \ln(p_i^u),\ y_{2i}^l = \ln(1-p_i^u), \text{ and } y_{2i}^u = \ln(1-p_i^l), \quad \forall i = 1, \ldots, I. \tag{3b}$$

Next, to linearize (2a) itself, denote

$$z_j = \ln(\theta_j), \quad \forall j \in \tau, \tag{4a}$$

where, based on (2b), we can impose

$$z_j^l \le z_j \le z_j^u,$$

$$\text{where } z_j^l = \ln(\theta_j^l) \text{ and } z_j^u = \ln(\theta_j^u), \quad \forall j \in \tau. \tag{4b}$$

Likewise, to accommodate the term $\ln(l_j)$ generated by taking logarithms in (2a), define

$$\xi_j = \ln(l_j), \quad \forall j \in \tau \tag{5a}$$

and impose the related bounds

$$\xi_j^l \le \xi_j \le \xi_j^u,$$

$$\text{where } \xi_j^l = \ln(l_j^l) \text{ and } \xi_j^u = \ln(l_j^u), \quad \forall j \in \tau. \tag{5b}$$

This yields the following equivalently reformulated problem ETO(Ω), which is predicated on the

hyperrectangle Ω defined as

$$\Omega \equiv \{(p, l, \theta): p_i^l \le p_i \le p_i^u, \forall i,\ l_j^l \le l_j \le l_j^u, \forall j,\ \theta_j^l \le \theta_j \le \theta_j^u, \forall j\}. \quad (6)$$

$$\textbf{ETO}(\Omega): \quad \text{Minimize} \sum_{j\in\tau} \theta_j \quad (7a)$$

subject to

$$\sum_{i=1}^{I} q_{im} \le s_m, \quad \forall m = 1, \ldots, M \quad (7b)$$

$$\sum_{j\in\tau} r_{jn} \le t_n, \quad \forall n = 1, \ldots, N \quad (7c)$$

$$\sum_{i=1}^{I}\sum_{m=1}^{M} c_{im} q_{im} + \sum_{j\in\tau}\sum_{n=1}^{N} d_{jn} r_{jn} \le \beta, \quad (7d)$$

$$y_{1i} - y_{2i} = a_{i0} - \sum_{m=1}^{M} a_{im} q_{im}, \quad \forall i = 1, \ldots, I \quad (7e)$$

$$l_j = b_{j0} - \sum_{n=1}^{N} b_{jn} r_{jn}, \quad \forall j \in \tau \quad (7f)$$

$$z_j = \xi_j + \sum_{k\in S_{1j}} y_{1k} + \sum_{k\in S_{2j}} y_{2k}, \quad \forall j \in \tau \quad (7g)$$

$$y_{1i} = \ln(p_i), \quad \forall i = 1, \ldots, I \quad (7h)$$

$$y_{2i} = \ln(1 - p_i), \quad \forall i = 1, \ldots, I \quad (7i)$$

$$z_j = \ln(\theta_j), \quad \forall j \in \tau \quad (7j)$$

$$\xi_j = \ln(l_j), \quad \forall j \in \tau \quad (7k)$$

$$(p, l, \theta) \in \Omega, \quad (7l)$$

$$(q, r) \in \mathcal{P}, \quad (7m)$$

$$\left\{ \begin{array}{l} z_j^l \le z_j \le z_j^u \text{ and } \xi_j^l \le \xi_j \le \xi_j^u, \quad \forall j \in \tau, \\ y_{1i}^l \le y_{1i} \le y_{1i}^u \text{ and } y_{2i}^l \le y_{2i} \le y_{2i}^u, \quad \forall i = 1, \ldots, I \end{array} \right\}. \quad (7n)$$

For the sake of convenience in the sequel, we shall denote the set of variables in Problem ETO(Ω), with obvious vector notation, as

$$x \equiv (p, l, q, r, \theta, y_1, y_2, z, \xi),$$

where, in particular, $y_1 \equiv (y_{1i}, \forall i)$ and $y_2 \equiv (y_{2i}, \forall i)$.

Observe that ETO(Ω) is, in general, a factorable program as analyzed by Sherali and Wang (2001), who have derived polynomial programming approximations and integrated this with the *reformulation-linearization technique* (RLT) of Sherali and Tuncbilek (1992) in order to solve such problems. However, in our case, ETO(Ω) is linear except for the complicating identities (7h)–(7k), where the latter are all defined in terms of univariate, monotone logarithmic functions. Hence, in lieu of adopting an RLT approach, we shall devise a specialized polyhedral outer-approximation mechanism, as explained below, to generate a linear programming relaxation (LP(Ω)) for computing lower bounds in a branch-and-bound framework. Moreover, this specialized scheme will enable us to conduct a partitioning of intervals for only a subset of variables.

Toward this end, consider the following relationship in generic notation, which can then be applied to each of the identities in (7h)–(7k):

$$\lambda = \ln(\gamma), \quad \text{where } 0 < \gamma^l \le \gamma \le \gamma^u < \infty. \quad (8)$$

We construct an associated polyhedral outer approximation for (8) via the affine convex envelope of this concave function, along with four tangential supports (equispaced for the sake of simplicity), including, in particular, supports at the endpoints γ^l and γ^u. This yields the following:

$$\lambda \ge \ln(\gamma^l) + \frac{\gamma - \gamma^l}{\gamma^u - \gamma^l}[\ln(\gamma^u) - \ln(\gamma^l)], \quad (9a)$$

along with

$$\lambda \le \ln(\gamma_w) + \frac{\gamma - \gamma_w}{\gamma_w},$$

$$\text{for } \gamma_w = \gamma^l + \frac{\gamma^u - \gamma^l}{3}(w - 1),\ w = 1, \ldots, 4. \quad (9b)$$

Hence, let LP(Ω) denote the linear program obtained from ETO(Ω) by replacing (7h)–(7k) with the approximating constraints of the type (9a) and (9b). Then, the following results are evident, where ν[P] denotes the optimal value for any optimization problem P.

Lemma 1. *ν[LP(Ω)] provides a lower bound on ν[ETO(Ω)]. Moreover, if $\bar{x}$ solves LP(Ω) and if this solution satisfies the constraints* (7h)–(7k)*, then it is also optimal to Problem ETO(Ω) with the same objective value.*

Proof. Obvious from construction. □

Lemma 2. *Let $(\bar{l}, \bar{q}, \bar{r})$ be part of an optimal solution to LP(Ω). Compute $\hat{p}$ via*

$$\hat{p}_i = \frac{\bar{g}_i}{1 + \bar{g}_i}, \quad \forall i, \quad \text{where } \bar{g}_i \equiv e^{[a_{i0} - \sum_{m=1}^{M} a_{im}\bar{q}_{im}]}, \forall i. \quad (10)$$

Then $(\hat{p}, \bar{l}, \bar{q}, \bar{r})$ is a feasible solution for Problem ETO and provides an upper bound $\overline{UB}$ for this problem as given by

$$\overline{UB} = \sum_{j\in\tau} \bar{l}_j \prod_{k\in S_{1j}} \hat{p}_k \prod_{k\in S_{2j}} (1 - \hat{p}_k). \quad (11)$$

Proof. Given $\bar{q}$, the value $\hat{p}$ via (10) satisfies (1e), and by virtue of (3b), (7e), and (7n), we also have that $p_i^l \le \hat{p}_i \le p_i^u$, $\forall i$. All the other constraints of Problem ETO are directly included within Problem LP(Ω), and so $(\hat{p}, \bar{l}, \bar{q}, \bar{r})$ is a feasible solution. □

A key result that enables us to design the specialized branch-and-bound algorithm proposed below and induces its convergence to (global) optimality is presented next.

LEMMA 3. *Let $\bar{x}$ be an optimal solution to Problem LP(Ω) having an objective function value ν[LP(Ω)]. Furthermore, suppose that each component of $(\bar{p}, \bar{l}, \bar{\theta})$ equals its corresponding lower and upper bounds. Then $\bar{x}$ also solves Problem ETO(Ω) with the same objective value.*

PROOF. Noting the construction of Problem LP(Ω) from Problem ETO(Ω), it is sufficient to show that the given conditions of the lemma imply that $\bar{x}$ satisfies (7h)–(7k). Toward this end, consider the generic case of the identity (8) that has been approximated by the relationships (9a) and (9b). It is easy to verify that if γ equals γ^l or γ^u, then (9a) and (9b) together imply that $\lambda = \ln(\gamma)$. Consequently, it directly follows that $\bar{x}$ satisfies (7h)–(7k). □

COROLLARY 1. *Let $\bar{x}$ be an optimal solution to LP(Ω). If $\bar{p}_i \in \{p_i^l, p_i^u\}$ for any $i \in \{1, \ldots, I\}$, then $\bar{y}_{1i} = \ln(\bar{p}_i)$ and $\bar{y}_{2i} = \ln(1 - \bar{p}_i)$. Likewise, for any $j \in \tau$, if $\bar{\theta}_j \in \{\theta_j^l, \theta_j^u\}$, then $\bar{z}_j = \ln(\bar{\theta}_j)$, and if $\bar{l}_j \in \{l_j^l, l_j^u\}$, then $\bar{\xi}_j = \ln(\bar{l}_j)$.*

PROOF. Evident from the proof of Lemma 3. □

3. A Global Optimization Branch-and-Bound Algorithm

In our computations, when we applied the commercial package BARON (version 7.5) (see Sahinidis 1996) to solve Problem ETO versus Problem ETO(Ω), we obtained significantly different results (see §4 for more details). Hence, to validate the results, we present in this section a specialized global optimization procedure. This proposed algorithm is motivated by Lemma 3 and Corollary 1, and it solves Problem ETO via ETO(Ω) and its relaxation LP(Ω). In this approach, we adopt a branch-and-bound process based on partitioning the hyperrectangle Ω. Consider any node subproblem ETO(Ω) associated with a particular Ω, where the bounds on the variables (y_1, y_2, z, ξ) are correspondingly given by (3b), (4b), and (5b), respectively, and where, given any bounds on (p, l), we adjust the specified bounds on θ based on (2b) according to

$$\theta_j^l \leftarrow \max\left\{\theta_j^l, l_j^l \prod_{k\in S_{1j}} p_k^l \prod_{k\in S_{2j}} (1 - p_k^u)\right\}, \quad \forall j \quad \text{and}$$

$$\theta_j^u \leftarrow \min\left\{\theta_j^u, l_j^u \prod_{k\in S_{1j}} p_k^u \prod_{k\in S_{2j}} (1 - p_k^l)\right\}, \quad \forall j.$$

Initially, the bounds on θ are taken as given by (2b). We first construct the relaxation LP(Ω), and solve this linear program to obtain an optimal solution $\bar{x}$. If the condition of Lemma 1 holds true, then we will also have solved the node subproblem ETO(Ω). Otherwise, we apply Lemma 2 to possibly update the incumbent solution, and as necessary, we branch at this node by partitioning Ω as follows.

Branching Rule. Find the most discrepant identity among (7h)–(7k) at the current relaxation solution $\bar{x}$ according to

$$\max\{|\bar{y}_{1i} - \ln(\bar{p}_i)|, |\bar{y}_{2i} - \ln(1 - \bar{p}_i)|, |\bar{z}_j - \ln(\bar{\theta}_j)|, |\bar{\xi}_j - \ln(\bar{l}_j)|, \forall i, j\}, \tag{12}$$

where ties are broken by favoring the term involving the log of the variable having the largest bounding interval length. If this identified maximum term is one of the first two types of terms, then we partition the interval of the corresponding variable p_i into two subintervals, $[p_i^l, \bar{p}_i]$ and $[\bar{p}_i, p_i^u]$, and if it is the third or the fourth type of term, we respectively branch on the corresponding variable θ_j or l_j by cutting its interval at the value $\bar{\theta}_j$ or $\bar{l}_j$.

At each stage s of this procedure, $s = 0, 1, 2, \ldots$, we will have a set of nonfathomed or *active nodes* A_s, where each node $a \in A_s$ is indexed by some hyperrectangle Ω^a. For each node $a \in A_s$, a lower bound LB_a is given by $\nu[\text{LP}(\Omega^a)]$, so that $\text{LB}(s) \equiv$ minimum $\{\text{LB}_a\colon a \in A_s\}$ is a *global lower bound* at stage s for Problem ETO. Whenever any lower-bounding node subproblem is solved, we apply Lemma 2, to possibly update the global upper bound ν^* on the overall problem ETO. Whenever $\text{LB}_a \geq \nu^*(1 - \varepsilon)$ for some optimality tolerance $0 \leq \varepsilon < 1$, we fathom node a. At any stage s, if $A_s = \varnothing$, then we stop with the incumbent solution as (ε-) optimal. Otherwise, we select an active node $a(s) \in \arg\min\{\text{LB}_{\hat{a}}\colon \hat{a} \in A_s\}$, partition it according to the foregoing branching rule, and continue.

THEOREM 1. *The proposed algorithm (run with $\varepsilon \equiv 0$) either terminates finitely with the incumbent solution being optimal to Problem ETO, or else an infinite sequence of stages is generated such that along any infinite branch of the branch-and-bound tree, any accumulation point of the (p, l, q, r)-variable part of the sequence of linear programming relaxation solutions generated for the corresponding node subproblems solves Problem ETO.*

PROOF. The case of finite termination is clear. Hence, suppose that an infinite sequence of stages is generated. Consider any infinite branch of the branch-and-bound tree generated via the sequence of nested hyperrectangles $\Omega^{a(s)}$ that correspond to a set of stages s in some index set S. Hence, we have

$$\text{LB}(s) = \text{LB}_{a(s)} = \nu[\text{LP}(\Omega^{a(s)})], \quad \forall s \in S. \tag{13}$$

For each node $a(s)$, $s \in S$, let $x^{a(s)}$ be the solution obtained for LP($\Omega^{a(s)}$). By taking any convergent subsequence, if necessary, using the compactness of the

feasible region, assume without loss of generality that $\{x^{a(s)}, \Omega^{a(s)}\}_S \to (x^*, \Omega^*)$. We must show that the (p^*, l^*, q^*, r^*)-variable part of the solution x^* solves Problem ETO. First, note that because $\text{LB}_{a(s)}$ in (13) is the least lower bound at stage s, we have

$$V^* \equiv \lim_{s\to\infty,\, s\in S} \text{LB}_{a(s)} \le \nu[\text{ETO}]. \tag{14}$$

Now, along this infinite branch of the enumeration tree, some p_i, θ_j, or l_j variable is partitioned infinitely, often over nodes $s \in S_1 \subseteq S$, say. Furthermore, by virtue of the partitioning scheme, it is evident that in the limit, this variable equals one of its bounds. Hence, from Corollary 1, its discrepancy in (12) approaches zero. However, because the discrepancy of this variable was the maximum in (12) for all $s \in S_1$, we have by taking limits as $s \to \infty, s \in S_1$, that x^* satisfies (7h)–(7k). Therefore, by Lemma 1, x^* solves ETO(Ω^*) with objective value V^*, which yields $\nu[\text{ETO}] \le V^*$. From (14), we then get $V^* = \nu[\text{ETO}]$, i.e., x^* solves Problem ETO. □

4. Computational Case Study

In this section, we present some computational results pertaining to a hypothetical case study related to a gasline rupture (see Figure 1) for various parameter settings, as well for several other simulated test cases. These results are, respectively, presented in §§4.1 and 4.2. All computations have been performed on a Solaris 500 MHz machine running on Sun OS 5.8, and using CPLEX 10.0 to solve the underlying LP relaxation lower-bounding problems.

4.1. Gasline Rupture Case Study

Suppose that some five event-related resources pertaining to investments in the four safety measures identified in Figure 1, and five consequence-related resource investments pertaining to appropriate environmental treatment and emergency response management strategies, are available for deployment to mitigate this particular hazard. We refer the reader to Desai (2005) for data pertaining to $\{a_{im}, b_{jn}, c_{im}, d_{jn}, l_j^l, l_j^u\}$-values. Moreover, we set $\beta = 7{,}500$, $s_m = 10$, $\forall\, m$, $t_n = 50$, $\forall\, n$, and $p_i^l = 0.0001$ and $p_i^u = 0.01$, $\forall\, i$.

The associated problem ETO(Ω), given by (7a)–(7n), was solved via the proposed branch-and-bound algorithm described in §3. The objective function value obtained at optimality is given by 80.9983. This is reduced from the total risk value of 1,364.5 corresponding to the no-action alternative (case of $q = 0$ and $r = 0$). Using $\varepsilon = 10^{-6}$, the proposed algorithm enumerated a total of 51 nodes over a CPU time of 0.13 seconds. From the optimal allocations (see Desai 2005), it was observed that using the available resources, the failure probabilities associated with four of the eight event points were at their upper bounds, but the others were considerably reduced in a strategic manner, in concert with judiciously mitigating the loss values associated with the end nodes.

For the purpose of comparison, Problem ETO, given by (1a)–(1h), was also directly solved by using the commercial global optimizer BARON (version 7.5) using default parameter settings (see Sahinidis 1996). In addition, we applied BARON v7.5 to an alternative nonlinear version of Problem ETO(Ω), in which θ_j is eliminated by equivalently writing (7a) as the convex objective function: Minimize $\sum_{j\in\tau} e^{z_j}$. Both of these computational experiments consumed 0.16 seconds and yielded identical results, and the best objective function value obtained was 92.9335, which is greater (worse) than the global optimum discovered by our algorithm by 14.73%. The solution to Problem ETO using BARON yielded quite different resource allocations, especially for the consequence-related alternatives, than those obtained by our proposed algorithm (see Desai 2005). In another computational experiment, we also solved our reformulated model, ETO(Ω), using BARON. In this case, BARON yielded the identical global optimum obtained by our proposed algorithms in 0.09 seconds of CPU time.

Finally, we examined the sensitivity of the model with respect to the consequence-related resource and budget availabilities, given, respectively, by $t_n \equiv t$, say, and β. (A preliminary computational experiment revealed that these two parameters had the most significant impact on the optimal objective function value.) Figure 2 displays the variation in the objective function when $t \in \{10, 30, 50, 70\}$ and $\beta \in \{2{,}500, 5{,}000, 7{,}500\}$. As expected, the objective function value follows the law of diminishing marginal returns, decreasing rapidly initially as the resource quantity resource increases, but flattening out for larger values of these parameters. Moreover, no further improvement for any t-value was obtained by increasing β beyond 7,500.

Figure 2 **Sensitivity of the Objective Function with Respect to the Parameters t and β**

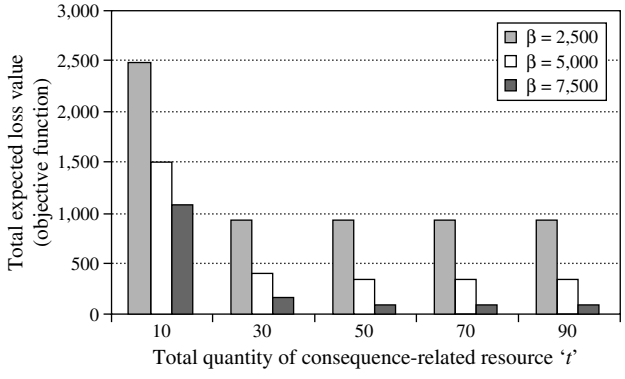


Table 1 Performance of the Proposed Algorithm and BARON for Simulated Test Instances

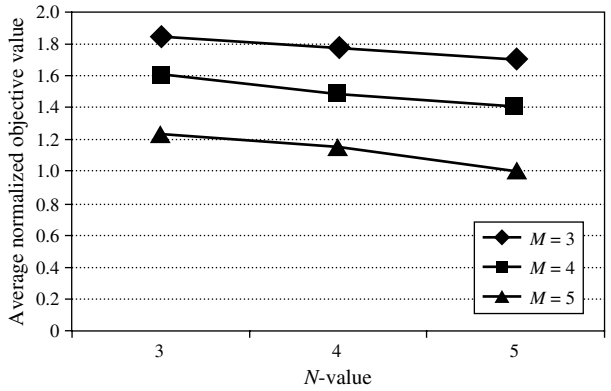
Problem instance	Problem size (no. of end nodes)	No. of variables	Proposed algorithm: No. of nodes enumerated	Proposed algorithm: CPU time (s)	BARON CPU time for solving ETO(Ω) (s)	BARON CPU time for solving ETO (s)	Z^*_{BARON}/Z^*
1	33	425	59	1.724	1.55	1.64	1.13
2	65	841	290	48.90	34.06	51.0	1.09
3	129	1,673	2,412	1,733.2	1,210.2	1,343.5	1.19
4	257	3,337	5,320	4,400.0	5,400.66	4,012.6	1.23
5	513	6,665	13,445	13,010.5	13,837.8	5,900.44	1.28
6	1,025	13,321	41,067	27,756	28,775.55	6,124.37	1.28

4.2. Simulated Test Cases

To further demonstrate the efficacy of the proposed algorithm, we tested its performance on several synthetically generated test cases, beginning with six instances where the number of end nodes in the binary event tree varied from $2^5 + 1 \equiv 33$ to $2^{10} + 1 \equiv 1{,}025$. For this first set of six test instances, we fixed $M = N = 5$, and we generated the elements c_{im} and d_{jn} of the cost matrices arbitrarily, but consistent with their physical interpretation, in the range $[20, 40]$, representing hourly costs, say. The logit model coefficients, a_{im}-values, $\forall (i, m)$, and b_{jn}-values, $\forall (j, n)$, were likewise generated over the intervals [0.5,1] and $[1{,}000, 5{,}000]$, respectively, and we set the total budget equal to $\$750|\tau|$. Note that all problems were initially tested for feasibility, and only feasible problems with nontrivial solutions were utilized for comparison purposes.

Table 1 displays the results obtained using the proposed branch-and-bound algorithm and BARON to solve Problem ETO(Ω), as well as implementing BARON v7.5 on the original model ETO for the sake of assessing the utility of the reformulated model ETO(Ω), where Z^* and Z^*_{BARON} denote the optimal value obtained by the proposed algorithm and that obtained by BARON applied to Problem ETO, respectively. From the tabulated results, it can be seen that the proposed algorithm consistently performed better than the direct solution of ETO via BARON, finding optimal solutions that are at an average 16.6% better in objective function value than the best solutions detected by BARON. Noting the increased computational effort required for the largest two instances, we also reran our algorithm using a 3% optimality tolerance limit, which yielded an average reduction of 17.34% in computational time required, while leading to a solution that was on average inferior by 2.64%. On the other hand, when BARON was applied to solve the reformulated model ETO(Ω), it produced identical solutions while consuming at an average 179.3 fewer CPU seconds (a decrease of 30.1%) for the first three relatively smaller instances, but an additional 949.2 CPU seconds (an increase of 6%) at an average for the next three relatively larger instances. (A t-test performed at a 95% confidence level revealed no statistically significant difference in the mean computational times of the proposed algorithms and BARON.)

Figure 3 Contour Plot Displaying the Average Normalized Objective Function Values



In another computational experiment, we modified the above six test instances by varying N and M from three to five. Figure 3 displays the ratio of the optimal objective function value obtained for each of these test instances to that for the case of $M = N = 5$, averaged over the six instances. Note that this normalized value is lower (better) for the cases having relatively higher values of M and lower values of N, in comparison to the cases having lower values of M and higher values of N. (Actually, this feature is accentuated sharply as the problem size increases.) Evidently, investing in event-related resources is better from a system safety perspective, as opposed to investing in risk control at a later stage (end nodes).

5. Summary and Conclusions

The contribution in this paper has focused on presenting a novel analytical modeling and algorithmic approach for strategically manipulating event probabilities and end-effect consequences through technological and emergency response investments in order to minimize the overall risk. This risk management problem was modeled as a nonconvex program and

was solved through a suitable variable transformation and polyhedral outer-approximation technique applied in concert with a specialized branch-and-bound procedure. The proposed algorithm was proven to converge to a global optimal solution. Computational results revealed the efficacy of using the reformulated model and solution approach, particularly in contrast with applying the commercial software BARON v7.5 to the original model formulation.

Several variations and extensions of this work can be considered for future research. For example, instead of minimizing the total expected loss, we could alternatively minimize the maximum component of this expected loss over the end nodes of the event tree. Variable transformation strategies identical to those adopted in reformulating Problem ETO could be employed to solve this problem. Also, the alternative model alluded to in §1 concerning exponential loss mitigation functions for end nodes could be investigated for both the expected and minimax objective functions. The performance of the proposed approach for other case studies where the event tree has general multistate outcomes, as opposed to a Bernoulli mechanism, is of interest. Finally, this modeling and algorithmic approach can be applied in the context of more general decision tree problems.

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